Figure 7.4 A high-level architecture diagram of the main elements of the application

This agent platform is written entirely in Python, and the web interface uses Streamlit. In the next section, we look at how to build an OpenAI LLM chat application.

7.2 Introducing Streamlit for chat application development

Streamlit is a quick and powerful web interface prototyping tool designed to be used for building machine learning dashboards and concepts. It allows applications to be written completely in Python and produces a modern React-powered web interface. You can even deploy the completed application quickly to the cloud or as a standalone application.

7.2.1 Building a Streamlit chat application

Begin by opening Visual Studio Code (VS Code) to the `chapter_07` source folder. If you've completed the previous exercise, you should already be ready. As always, if you need assistance setting up your environment and tools, refer to appendix B.

We'll start by opening the `chatgpt_clone_response.py` file in VS Code. The top section of the code is shown in listing 7.3. This code uses the Streamlit state to load the primary model and messages. Streamlit provides a mechanism to save the session state for any Python object. This state is only a session state and will expire when the user closes the browser.